Exploratory Data Analysis: E-commerce Orders

Executive Summary | DecodeLabs Data Analytics Internship, Project 2

Prepared by Muhammad Waqar Tahir | 1,200 orders, 2023-01-01 to 2025-06-30

Bottom line

1. **Reported revenue is well above real revenue.** 41% of order value is cancelled or returned, and only 39% is delivered or shipped.
2. **The order-status data cannot be trusted.** 218 of 237 pending orders (92%) are more than 90 days old, holding about 235,000 of value.
3. **Nothing in the data explains why orders fail.** No product, payment method, channel or coupon differs meaningfully, so the business needs to start recording cancellation and return reasons.

1. Problem statement

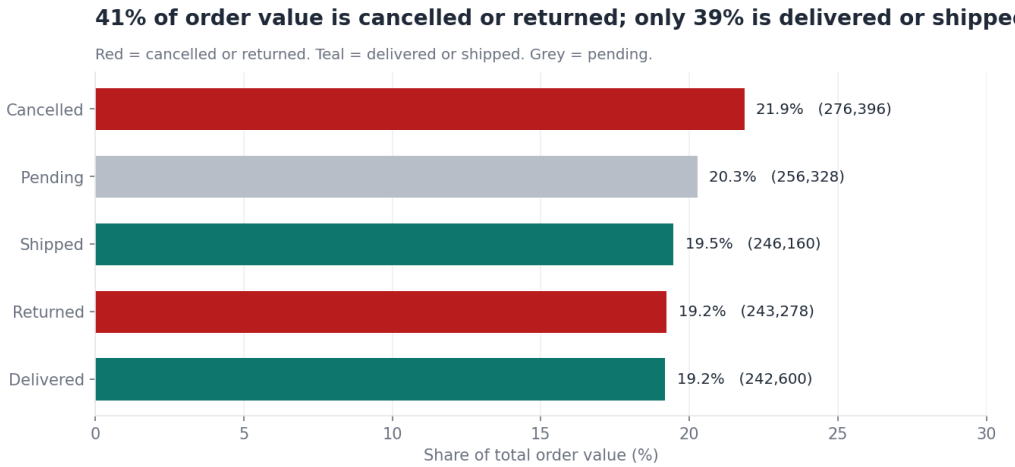
Where is this business gaining or losing value, and which patterns in the order data can be trusted enough to act on? The analysis covers order value, outliers, time trends, order status and the relationships between variables.

2. Methodology

The input is the cleaned dataset from Project 1 (1,200 orders, no missing values, no duplicate IDs). Analysis used Python (pandas, SciPy, Matplotlib): descriptive statistics (count, mean, median, five-number summary), outlier detection by both the IQR and Z-score methods, correlation analysis, and comparisons across product, payment method, referral source and coupon. **All 14 statistical tests were run as one family and corrected for multiple testing (Holm-Bonferroni)**, so a difference is only called real if it survives the correction. Order age is measured against the latest order date in the data, not today's date.

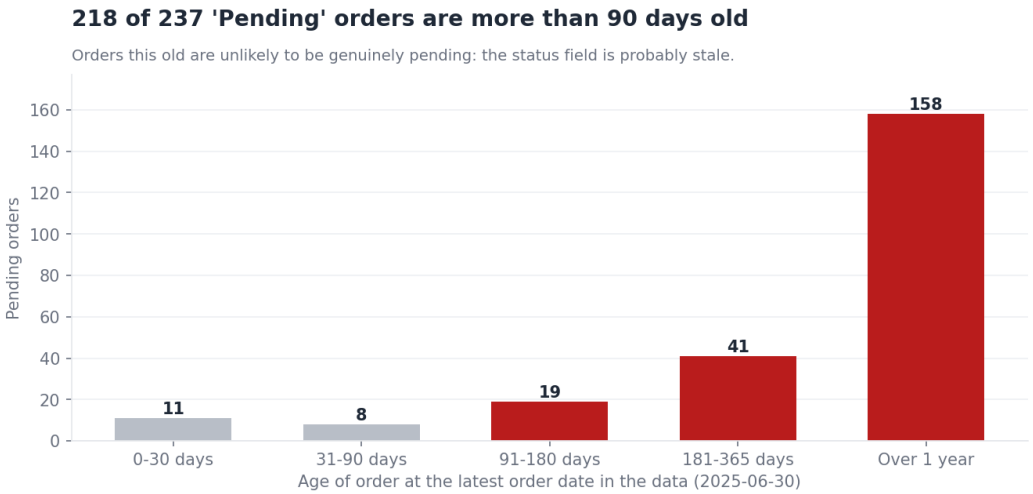
3. Key findings

FINDING 1



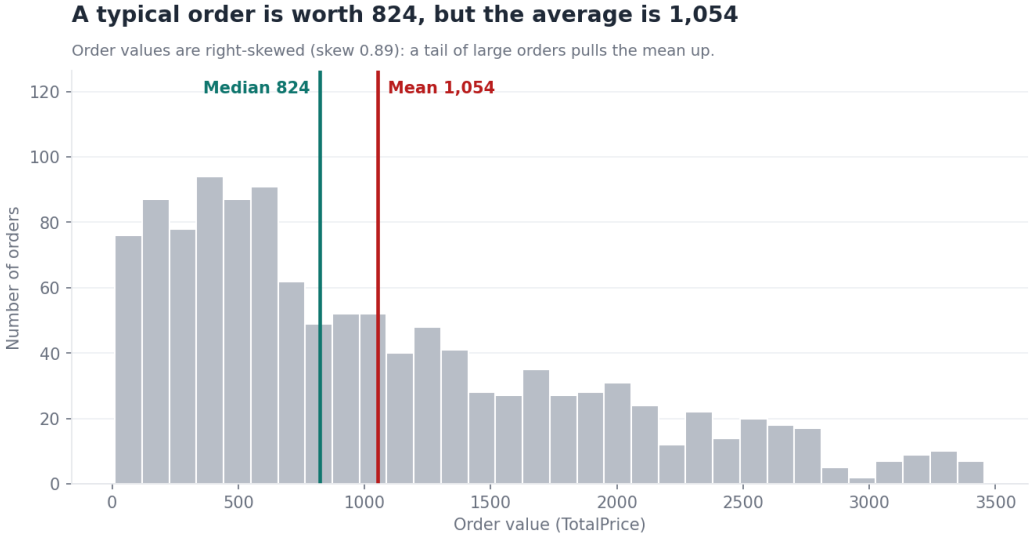
Of 1,264,762 in gross order value, 519,674 sits in cancelled or returned orders (497 orders, 41.4%). Gross order value is about 2.6 times the value actually delivered or shipped, so any revenue figure that adds up every row overstates what has been fulfilled.

FINDING 2



Of these, 158 are more than a year old. Orders do not stay genuinely pending that long, so the status field is probably stale or orders were abandoned and never closed. About 235,000 of value is sitting in a status nobody can rely on. This is a process problem the business can check and fix, which makes it the most actionable finding.

FINDING 3



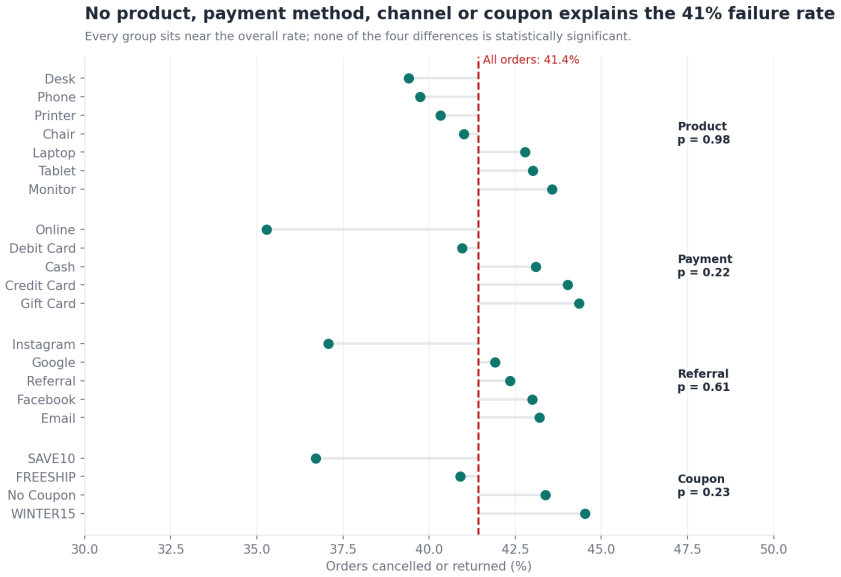
Order value is right-skewed (skew 0.89), so the mean overstates a typical order by about 28%. The largest 20% of orders carry 45.5% of all value. Middle half of orders: 411 to 1,578.

FINDING 4



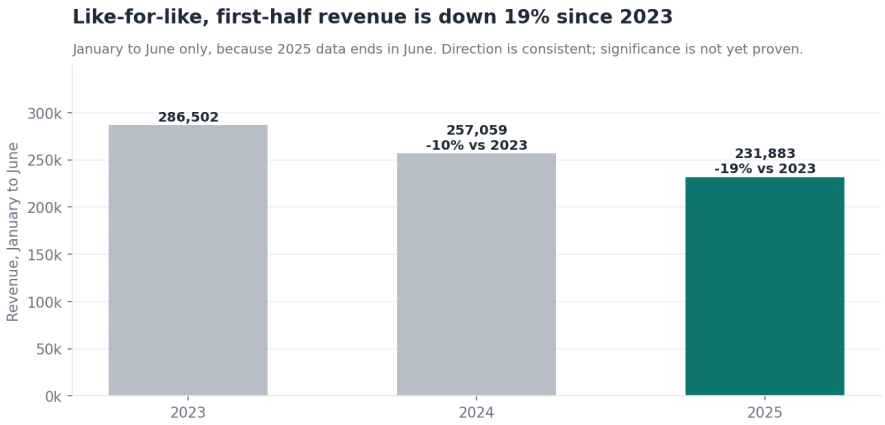
The IQR method flags 8 orders above 3,330; the Z-score method flags 0 because skew inflates the standard deviation (the largest order has Z = 2.93). All 8 are maximum-quantity orders (5 units) at unit prices of 667 or more. They are legitimate, so they were **kept, not deleted**. Together they are 2.1% of value; 4 of the 8 were cancelled or returned.

FINDING 5



Failure rates range from about 35% to 45% across groups, but none of the four comparisons is statistically significant (p = 0.22 to 0.98). Coupons are used on 74% of orders, yet median order value and failure rate are essentially the same with and without one. The cause is not visible in the current data.

FINDING 6



January to June revenue fell 286,502 to 257,059 (-10%) to 231,883 (-19%). The direction is consistent, but monthly orders vary a lot (mean 40, standard deviation 7.7), and the trend across all 30 months is not statistically significant (revenue p = 0.11, orders p = 0.34). Treat it as a watch item, not an alarm.

FINDING 7

Correlations here are arithmetic, not behaviour

Order value correlates with quantity (0.61) and unit price (0.72) only because TotalPrice is defined as Quantity x UnitPrice. Quantity and UnitPrice are unrelated (r = 0.01). Quantity can never exceed the items in the cart, which explains their 0.65 correlation. None of this reveals customer behaviour, and correlation would not prove cause and effect if it did.

A note on rigour. 14 tests were run. Before correction, 1 looked significant: whether the full status mix differs by referral source (raw p = 0.005). After correcting for 14 tests its adjusted p-value is 0.073, above the 0.05 line, which is what one lucky result among many looks like. It is kept as a hypothesis to re-test on new data, not reported as a finding. After correction, 0 of 14 tests are significant.

4. What it means for the business

What the data says	What it means
Median order is 824, mean is 1,054; top 20% of orders carry 45% of value	Use the **median** for a typical customer. The business leans on a minority of larger orders.
41.4% of orders fail; 38.6% of value is delivered or shipped	Fulfilment is the **largest lever** in the data. Reported revenue overstates real revenue.
218 pending orders are over 90 days old	The status process is broken or unmonitored. A **process fix** would clean up the picture.
No segment explains the failure rate	The answer needs **data not yet collected** (cancellation and return reasons).
Coupons show no measurable lift in order value or completion	Discounting is **not demonstrably paying for itself**; the file cannot show its cost.

5. Recommendations

#	Action	Evidence
1	**Audit the order-status process.** Review the stale pending orders, close or cancel the dead ones, and add an alert for orders pending beyond about 30 days.	218 orders, about 235,000 in value
2	**Record why orders fail.** Add cancellation and return reason codes and status-change timestamps.	41.4% failure rate is unexplained
3	**Report the median beside the mean**, and use IQR rather than Z-score to flag unusual orders.	Mean overstates a typical order by 28%
4	**Track monthly revenue on a control chart** and revisit after the second half of 2025.	First-half revenue -19% vs 2023, not yet significant
5	**Test coupons in a controlled way** before further discounting, and add the discount amount to the data.	74% of orders use a coupon; no measurable lift
6	**Follow up the 8 largest orders** as a group of high-value accounts.	4 of 8 were cancelled or returned

6. Limitations

The data appears synthetic or simplified: categories are almost perfectly evenly spread and every shipping address is a house number on 'Main St', so location cannot be analysed. TotalPrice ignores coupons and shipping, UnitPrice varies widely within a single product, no currency or cost data is provided, 2025 covers only six months, and the data is observational, so nothing here proves cause and effect. Numbers are shown without a currency for that reason.